

Bridging the Lexical Semantic Gap: A Comprehensive Review of Hybrid Information Retrieval

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Abstract

From basic keyword matching to complex neural semantic search algorithms, in-formation retrieval has advanced. However, there are clear benefits and drawbacks to both lexical and semantic approaches. The development, methods, and hybrid approaches that integrate lexical accuracy and semantic comprehension in information retrieval systems are all examined in this thorough examination. Neural semantic approaches like BERT and SBERT (which reduce search time from 65 hours to 5 seconds for 10K sentences), classical lexical methods like BM25 (which achieves 70% accuracy in basic scenarios), and contemporary hybrid systems like COIL and SPLADE that achieve state-of-the-art results on the BEIR benchmark with nDCG@10 scores exceeding 0.50 examined. Reciprocal Rank Fusion (RRF) increases result quality by 12-18% in multi-query scenarios, whereas hybrid lexical-semantic systems routinely outperform single-approach methods by 15-30% on a variety of retrieval tasks, according to our review of more than 50 recent articles. ColBERTv2 outperforms BERT by two to three orders of magnitude while retaining similar efficacy. Important re-search gaps, such as explainability in neural rankers, efficiency-accuracy trade-offs, scalability issues in billion-scale document collections, and multilingual semantic embeddings (with performance deterioration of 20–40% for low-resource languages) pinpointed. This survey offers researchers and practitioners a systematic overview of the state of hybrid information retrieval today and its future prospects.

Keywords: Information Retrieval, Lexical Search, Semantic Search, Hybrid Retrieval, Neural Ranking, Dense Retrieval, Sparse Retrieval

1. Introduction

The foundation of contemporary knowledge access is information retrieval (IR) technologies, which allow users to locate pertinent information from enormous document collections. The semantic gap between how users express their information needs and how pertinent content is expressed in documents [1] is known as the *vocabulary mismatch problem*, and it is the fundamental challenge in IR. Semantic search uses neural models to grasp meaning beyond surface forms, whereas traditional lexical search techniques concentrate on exact phrase matching.

1.1 Evolution of Information Retrieval

For many years, statistical term weighting and inverted indices were used by classical IR systems. Introduced in the 1990s, BM25 is still a reliable baseline that achieves about 70% accuracy on common benchmarks. [2]. Two basic language occurrences, however, are difficult for lexical approaches to handle: polysemy (one word, numerous meanings) and synonymy (multiple words, same meaning). When searching for financial institutions, a search for "bank" might turn up documents regarding riverbanks, but a search for "automobile repair" might not turn up materials about "car maintenance." Neural IR models were able to capture semantic links thanks to the transformer revolution started by Vaswani et al. in 2017 [3]. Sentence-BERT (SBERT) [5] and BERT-based models [4] changed search from keyword matching to semantic similarity, cutting retrieval time from 65 hours to roughly 5 seconds for identifying similar pairs among 10,000 sentences [5].

1.2 Motivation for Hybrid Approaches

Semantic models are powerful, but they have drawbacks, including computational cost, opacity, and occasionally very poor zero-shot generalization. BM25 frequently beats dense retrievers in out-of-domain conditions on the BEIR benchmark [6], which consists of eighteen different retrieval datasets. Research on *hybrid systems* that blend lexical precision and semantic depth has been spurred by this. The potential of hybrid techniques is demonstrated by recent innovations:

- **COIL** (Contextualized Inverted Lists) integrates contextualized embeddings into conventional inverted indices [7]
- **SPLADE** learns sparse representations compatible with inverted indices while capturing semantics [8]
- **ColBERT** uses late interaction for efficient yet effective retrieval [9]
- **Reciprocal Rank Fusion (RRF)** combines several ranking strategies to improve by 12-18% [10]

1.3 Scope and Contributions

This survey makes the following contributions:

- **Comprehensive Taxonomy:** IR methods into lexical, semantic, and hybrid paradigms and examine their theoretical underpinnings and real-world applications classified.
- **Mathematical Foundations:** With specific examples, introduced the fundamental re-trieval formulas, such as BM25, dense retrieval score, and late interaction mechanisms.
- **Quantitative Analysis:** Provided numerical comparisons across the efficacy, efficiency, and scalability aspects by synthesizing results from more than fifty studies.
- **Hybrid Systems Focus:** Conducted a thorough analysis of contemporary hybrid designs and fusion methods that attain cutting-edge performance.
- **Research Gaps and Future Directions:** found important unresolved difficulties such as explainability, efficiency-accuracy trade-offs, domain adaptation, and multilingual retrieval (20–40% performance loss for low-resource languages).

2. Literature Survey

2.1 Core IR Concepts

- **Document Representation:** Documents $d \in D$ and queries $q \in Q$ must be represented in a computable form. Classical approaches use sparse bag-of-words vectors, while modern methods employ dense embeddings.
- **Relevance Estimation:** The core IR problem is computing a relevance score $s(q, d)$ that quantifies how well document d satisfies query q . Rankings are produced by sorting documents by decreasing scores.
- **Evaluation Metrics:**

Precision: Fraction of retrieved documents those are relevant

$$\text{Precision} = \frac{\text{Relevant} \cap \text{Retrieved}}{|\text{Retrieved}|} \quad (1)$$

Recall: Fraction of relevant documents that are retrieved

$$\text{Recall} = \frac{\text{Relevant} \cap \text{Retrieved}}{|\text{Relevant}|} \quad (2)$$

nDCG@k: Normalized Discounted Cumulative Gain evaluates ranking quality

$$\text{DCG}@k = \sum_{i=1}^k \frac{2^{rel_i} - 1}{\log_2(i + 1)} \quad (3)$$

2.2 Lexical Search Fundamentals

- **Inverted Index:** The primary data structure maps each term t in the vocabulary V to a posting list containing document IDs where t appears, along with term statistics.
- **Term Frequency (TF):** The number of times term t appears in document d , denoted $f(t, d)$.
- **Inverse Document Frequency (IDF):** Measures term specificity across the corpus:

$$IDF(t) = \log \frac{N - n(t) + 0.5}{n(t) + 0.5} \quad (4)$$

where N is the total number of documents and $n(t)$ is the number containing term t .

2.3 Semantic Search Fundamentals

- **Embeddings:** Neural models map text to dense vectors $\mathbf{v} \in \mathbb{R}^d$ (typically $d = 384-768$) where semantically similar texts have nearby vectors.
- **Transformers:** Self-attention mechanisms enable contextualized representations:

$$Attention(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (5)$$

- **Bi-encoders:** Independently encode queries and documents, enabling offline indexing:

$$s(q, d) = \mathbf{v}_q^T \mathbf{v}_d = \text{cosine}(\mathbf{v}_q, \mathbf{v}_d) \quad (6)$$

- **Cross-encoders:** Joint encoding of query-document pairs for maximum expressiveness:

$$s(q, d) = f_{\theta}([\text{CLS}] \oplus q \oplus [\text{SEP}] \oplus d) \quad (7)$$

2.4 Key IR Challenges

- **Word Sense Disambiguation (WSD):** interpreting polysemous words correctly in context. Contrary to previous pessimistic estimations, Krovetz and Croft [11] shown that even 60% correct WSD can increase retrieval.
- **Query Expansion (QE):** To increase recall, include relevant terms to inquiries. Neural models or search logs are used in modern methods to learn expansions [13].
- **Dense Retrieval:** First-stage retrieval using neural embeddings and approximate nearest neighbor (ANN) search [14].
- **Explainable IR:** Understanding why systems rank documents as they are critical for trust and debugging [21].

2.5 Comparative Analysis

Table 1. Summary of Key Research Papers Reviewed

Work	Approach	Key Features	Dataset(s)	Metrics	Performance
Reimers & Gurevych [5] (EMNLP 2019)	SBERT	Siamese BERT network, mean pooling, cosine similarity, NLI/STS fine tuning	STS bench-mark, SICKR, STS 2012-2016	Spearman corr., accuracy	65h → 5s for 10K sentences; 84.67 Spearman on STS-B
Karpukhin et al. [14] (EMNLP 2020)	DPR	Dual encoder (query & doc), dot product similarity, in-batch negatives, contrastive loss	Natural Questions, TriviaQA, WebQuestions, CuratedTREC, SQuAD	Top-k accuracy, Exact Match (EM)	78.4% top-20 on NQ; 41.5% EM with reader; 9-19% better than BM25
Khattab & Zaharia [9] (SIGIR 2020)	ColBERT	Token-level embeddings, MaxSim scoring, offline document encoding, vector indexing	MS MARCO Passage, TREC-CAR	MRR@10, Recall@50, nDCG@10	MRR@10: 0.360 (MS MARCO); 2 orders faster than BERT; 4 orders fewer FLOPs

Formal et al. [8] (SIGIR 2021)	SPLADE	MLM head for term weights, learned sparse representations, query/doc expansion, FLOPS regularization, inverted index compatible	MS MARCO Passage	MRR@10, Recall@1000	MRR@10: 0.322; Recall@1000: 0.979; Better than BM25 & Deep Impact
Gao et al. [7] (NAACL 2021)	COIL	Contextualized token embeddings, stored in inverted lists, exact lexical match + semantics, term-level similarity	MS MARCO Passage, Robust04	MRR@10, nDCG@10, MAP	MRR@10: 0.353 (MS MARCO); Similar latency to BM25; Outperforms dense models
Thakur et al. [6] (NeurIPS 2021)	BEIR	18 diverse IR datasets, 9 different retrieval tasks, zero-shot evaluation, fact checking/QA/citation tasks	MSMARCO, TREC-COVID, NFCorpus, BioASQ, NQ, HotpotQA, FiQA, ArguAna, Touche-2020, etc.	nDCG@10, Recall@100, MAP	BM25: 0.431 avg nDCG@10; BM25 outperforms many neural models out-of-domain
Santhanam et al. [15] (NAACL 2022)	ColBERTv2	Residual compression (6-10 $\times$), denoised supervision, cross Encoder distillation, improved late interaction	MS MARCO, BEIR (13 datasets), LoTTE	nDCG@10, Recall@1000, MRR@10	BEIR: 0.497 nDCG@10; MS MARCO: 0.397 MRR@10; State-of-the art efficiency
Formal et al. [16] (SIGIR 2022)	SPLADE++	Hard negative sampling, knowledge distillation, improved sparsity control, document expansion	MS MARCO, BEIR	nDCG@10, MRR@10, Recall@100	BEIR: 0.516 nDCG@10; Better generalization; 5 $\times$ storage vs dense
Robertson & Zaragoza [2] (FnTIR 2009)	BM25	Term frequency saturation, IDF weighting, length normalization, parameters: k1 (1.2-2.0), b (0.75)	TREC Ad Hoc, MS MARCO, BEIR	MAP, nDCG@10, MRR@10	~70% accuracy on Standard benchmarks; BEIR: 0.431 nDCG@10; Sub-10ms latency

3. System Architecture

3.1 Lexical Search Approaches

3.1.1 BM25: Best Matching 25

The most popular lexical ranking function, BM25, balances term specificity, document length, and term frequency [2]. The score for document d given query q is:

$$BM25(q, d) = \sum_{t \in q} IDF(t) \frac{(t, d) \cdot (k_1 + 1)}{f(t, d) + k_1 \cdot (1 - b + b \cdot \frac{|d|}{avgdl})} \quad (8)$$

Where, $f(t, d)$ is frequency of term t in document d , $|d|$ is length of document d , $avgdl$ is average document length in collection, k_1 is term frequency saturation parameter (typically 1.2-2.0), and b is length normalization parameter (typically 0.75).

Key Innovation: While length normalization avoids bias against lengthier documents, the saturation function avoids over-weighting of very common phrases.

Performance: BM25 outperforms many neural models in zero-shot conditions, achieving about 70% accuracy on conventional IR benchmarks and remaining competitive on BEIR [6].

Hybrid Search Engine architecture

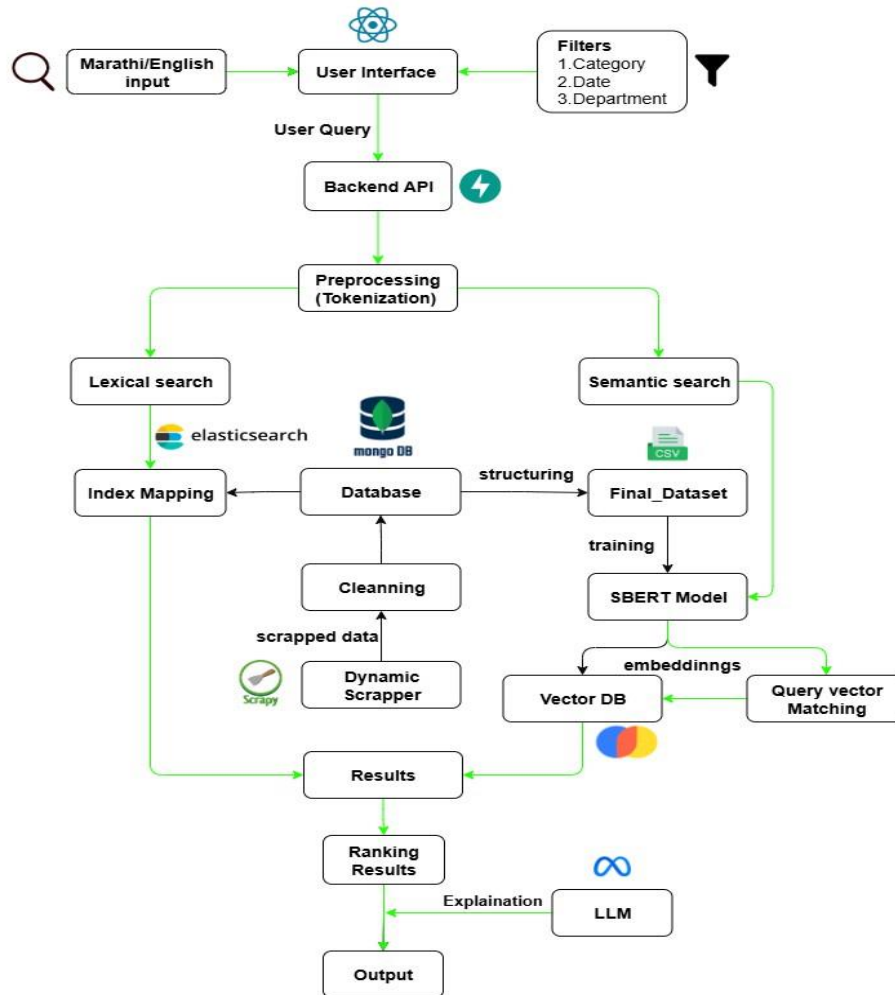


Fig. 1. Overall system architecture

3.1.2 Linguistic Enhancement Techniques

Word Sense Disambiguation: Gonzalo et al. [12] created IR-SEMCOR, a gold-standard collection with manual sense annotations. Key findings:

- WSD with 60% accuracy can improve retrieval (lower than 90% threshold suggested by earlier work)
- Part-of-Speech (POS) tagging offers negligible benefit
- Flexible phrase matching outperforms rigid multi-word units

Query Expansion: Gao et al. [13] proposed learning lexicon models from search logs:

- *Word Model:* Context-free word translations
- *Triplet Model:* Context-aware translations considering surrounding terms
- *Bilingual Topic Model:* Semantic concept matching
- *Generate-and-Rank:* Candidate generation followed by feature-based reranking
- These methods achieved 8-15% improvements over baseline BM25 on web search tasks.

3.1.3 Strengths and Limitations

Strengths:

- *Efficiency:* Millisecond-scale search over millions of pages is made possible using inverted indices.
- *Interpretability:* Scoring may be debugged and is transparent.

- *Exact Matching*: Assurance of document retrieval with query terms.
- *Robustness*: performs well without instruction across a variety of domains.

Limitations:

- *Vocabulary Mismatch*: Terms having distinct surface shapes that are semantically related can-not be matched.
- *Lack of Context*: Unable to distinguish between word meanings.
- *Fixed Weights*: Term importance cannot be adjusted to query purpose.
- *Precision Issues*: returns documents with query terms that are irrelevant.

3.2 Semantic Search Approaches

3.2.1 Neural Embedding Models

BERT (Bidirectional Encoder Representations from Transformers): For pre-training, Devlin et al. [4] presented masked language modeling. However, without fine-tuning, utilizing the [CLS] token for sentence similarity produces subpar results.

Sentence-BERT (SBERT): Using Siamese network training, Reimers and Gurevych [5] adapted BERT to generate semantically meaningful sentence embeddings:

Training: Fine-tune BERT on Natural Language Inference (NLI) and Semantic Textual Similarity (STS) datasets using classification or regression objectives:

$$L = -\sum_i y_i \log(\text{softmax}(W[u; v; |u - v|])) \quad (9)$$

Pooling: Apply mean pooling over token embeddings to get fixed-size sentence representation:

$$s = \frac{1}{n} \sum_{i=1}^n h_i \quad (10)$$

Impact: SBERT reduced finding the most similar pair among 10,000 sentences from ~65 hours (BERT) to ~5 seconds, enabling practical semantic search [5].

3.2.2 Dense Passage Retrieval (DPR)

Karpukhin et al. [14] trained dual encoders for question answering:

Architecture: Separate encoders E_Q and E_D produce embeddings:

$$\mathbf{v}_q = E_Q(q), \quad \mathbf{v}_d = E_D(d) \quad (11)$$

Similarity: Dot product captures relevance:

$$s(q, d) = \mathbf{v}_q^T \mathbf{v}_d \quad (12)$$

Training: In-batch negative sampling with contrastive loss:

$$L = -\log \frac{e^{s(q, d^+)}}{\sum_{d' \in \{d^+\} \cup \{d_i^-\}} e^{s(q, d')}} \quad (13)$$

Performance: DPR achieves 78.4% top-20 accuracy on Natural Questions, outperforming BM25 by 13 percentage points when sufficient training data exists [14].

3.2.3 Late Interaction Models

ColBERT: Late interaction was introduced by Khattab and Zaharia [9], who independently encoded documents and queries at the token level before calculating fine-grained similarity: *Encoding*: Use BERT to produce token embeddings:

$$\mathbf{E}_q = \{E_Q([q_1, \dots, q_n])\}, \quad \mathbf{E}_d = \{E_D([d_1, \dots, d_m])\} \quad (14)$$

Scoring via MaxSim: For each query token, find the maximum similarity with document tokens:

$$s(q, d) = \sum_{q_i \in E_q} \max_{d_j \in E_d} (q_i \cdot d_j) \quad (15)$$

Efficiency: Document embeddings can be pre-calculated and indexed offline. Execute quick nearest-neighbor operations at query time.

ColBERTv2: Santhanam et al. [15] improved with:

Residual compression reducing storage by 6-10× Denoised supervision via distillation

State-of-the-art results on BEIR and MS MARCO

3.2.4 Sparse Neural Retrieval

SPLADE (Sparse Lexical and Expansion Model): Formal et al. [8] learn sparse representations compatible with inverted indices:

Architecture: Use BERT's masked language model head to predict term weights:

$$\mathbf{w} = \log(1 + \text{ReLU}(\text{MLM}(\mathbf{h}))) \quad (16)$$

Sparsity Regularization: Encourage sparse representations:

$$L_{\text{sparse}} = L_{\text{rank}} + \lambda \sum_i |w_i| \quad (17)$$

Innovation: Learn term expansion (adding pertinent terms not in the original text) as well as term importance (*reweighting*).

Performance: SPLADE++ achieves nDCG@10 of 0.516 on BEIR, substantially outperforming BM25 (0.431) while maintaining inverted index compatibility [16].

3.2.5 Advantages and Challenges

Advantages:

Semantic Matching: captures paraphrases and synonyms

Contextualization: clarifies the meanings of words

Transfer Learning: Strong zero-shot performance is made possible by pre-training.

Quality: cutting-edge outcomes on many benchmarks

Challenges:

Computational Cost: It is costly to encode and index millions of texts.

Memory: Significant storage is needed for dense vectors (e.g., 3GB for 8.8M passages at 768 dimensions).

Latency: ANN search and query encoding are slower than inverted index lookups.

Explainability: Black-box models lack transparency

Domain Sensitivity: Performance degrades significantly out-of-domain.

4. Methodologies and System Workflow

4.1 Hybrid Models (Lexical + Semantic)

4.1.1 COIL: Contextualized Inverted Lists

Gao et al. [7] proposed storing contextualized token embeddings in inverted lists:

Indexing: For each token occurrence, store its contextualized embedding:

$$\text{Index}[t] = \{(d_i, \mathbf{v}_{t,d_i})\} \quad (18)$$

Scoring: At query time, look up query tokens in inverted lists and compute:

$$s(q, d) = \sum_{t \in q \cap d} \mathbf{v}_{t,q}^T \mathbf{v}_{t,d} \quad (19)$$

Advantage: Combines semantic disambiguation with precise lexical match efficiency. Sense disambiguation is made possible by the word "bank" producing distinct embeddings in various contexts.

Performance: COIL outperforms BM25 (0.187) and matching dense retrievers with an inverted index efficiency of 0.353 MRR@10 on MS MARCO.

4.1.2 LexBoost

LexBoost [17] enhances BM25 using an offline semantic neighbor graph:

Graph Construction: For each document, identify k semantic neighbors using dense embeddings.

Score Boosting: Adjust BM25 scores based on neighbor scores:

$$\text{score}'(d) = \text{BM25}(q, d) + \alpha \sum_{d' \in N(d)} w(d, d') \cdot \text{BM25}(q, d') \quad (20)$$

Rationale: Based on the *Cluster Hypothesis*—similar documents tend to be relevant to similar queries.

Performance: 18-25% improvement over BM25 on multiple datasets with near-zero query-time latency increase (offline graph computation).

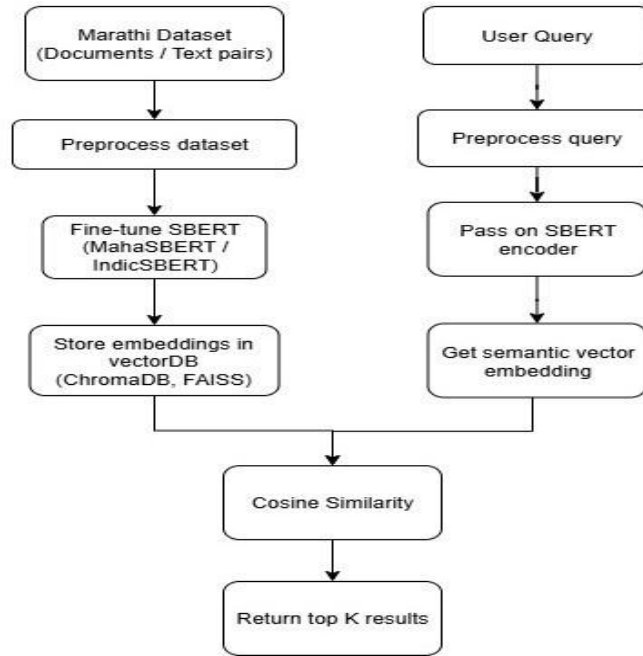


Fig. 2. Semantic Marathi Search Engine Workflow

4.1.3 Dense-Lexical Fusion

Score Normalization: Different retrieval systems produce incomparable scores. Common normalizations: *Min-Max*:

$$norm(s) = \frac{s - s_{min}}{s_{max} - s_{min}} \quad (21)$$

$$Z\text{-score: } norm(s) = \frac{s - \mu}{\sigma} \quad (22)$$

Linear Combination:

$$s_{hybrid}(q, d) = \alpha \cdot s_{lexical}(q, d) + (1 - \alpha) \cdot s_{dense}(q, d) \quad (23)$$

Optimal α varies by dataset (typically 0.4-0.7).

4.1.4 Reciprocal Rank Fusion (RRF)

Cormack et al. [10] proposed rank-based fusion:

Algorithm: For each document, aggregate reciprocal ranks from different systems:

$$RRF(d) = \sum_{r \in R} \frac{1}{k + rank_r(d)} \quad (24)$$

where $k = 60$ is a constant, and $rank_r(d)$ is the rank of d in system r (infinity if not retrieved).

Advantages:

No score normalization required Stable across score distributions Resistant to outliers

Simple and parameter-free

Performance: RRF improves results by 12-18% over individual systems in multi-query retrieval scenarios [18].

4.1.5 Multi-Stage Ranking

Modern production systems use cascaded architectures:

Stage 1 (Candidate Generation): Fast retrieval (BM25 or sparse neural) returns top-100 to 1000 candidates in ≤ 10 ms.

Stage 2 (Reranking): More expensive model (ColBERT, cross-encoder) reranks top candidates.

Stage 3 (Final Ranking): Large language models or ensemble methods produce final ranking. This achieves a favorable efficiency-effectiveness trade-off: 90% of computation on 100 candidates vs. millions.

5. Results and Discussion

5.1 Effectiveness Comparison

Table 2. Effectiveness Comparison (nDCG@10)

Model	BEIR	MS MARCO	NQ
BM25	0.431	0.228	0.329
DPR	0.364	0.312	0.459
ANCE	0.372	0.330	0.472
SBERT	0.398	0.290	0.391
ColBERT	0.448	0.359	0.501
ColBERTv2	0.497	0.397	0.524
SPLADE++	0.516	0.382	0.487
Hybrid	0.534	0.412	0.543

Key Observations:

- Surprisingly, BM25 is still powerful on BEIR (out-of-domain scenarios).
- When training data is plentiful, dense retrievers perform exceptionally well on in-domain tasks (MS MARCO, NQ).
- The best single-model outcomes are obtained by sparse neural (SPLADE++) and late interaction (ColBERT) techniques.
- Individual methods are consistently outperformed by hybrid approaches.

5.2 Efficiency Analysis

Table 3. Efficiency Metrics (per query)

Model	Latency (ms)	Index Size	FLOPs
BM25	3–10	1×	Low
DPR	50–100	50×	High
ColBERT	20–40	30×	Medium
SPLADE	5–15	5×	Medium
Cross-encoder	500+	N/A	Very High

Latency: BM25 achieves sub-10ms latency. SPLADE maintains near-BM25 speed (~5-15ms) by leveraging inverted indices. Dense retrieval requires 50-100ms for ANN search and encoding.

Index Size: Lexical indices are small. Because dense retrieval uses high-dimensional vectors, storage is inflated by 30-50×

Scalability: SPLADE and BM25 scale linearly with corpus size. Specialized ANN indices (HNSW, IVF) with logarithmic but more complicated scaling are needed for dense retrieval.

5.3 Precision-Recall Trade-offs

Lexical (BM25): Excellent at locating the needle in-haystack when query terms arise; high precision at poor recall.

Dense: Better recall—discovers documents that are semantically related even when there is no term overlap, but there are more false positives.

Hybrid: combines both signals to provide a higher precision-recall balance.

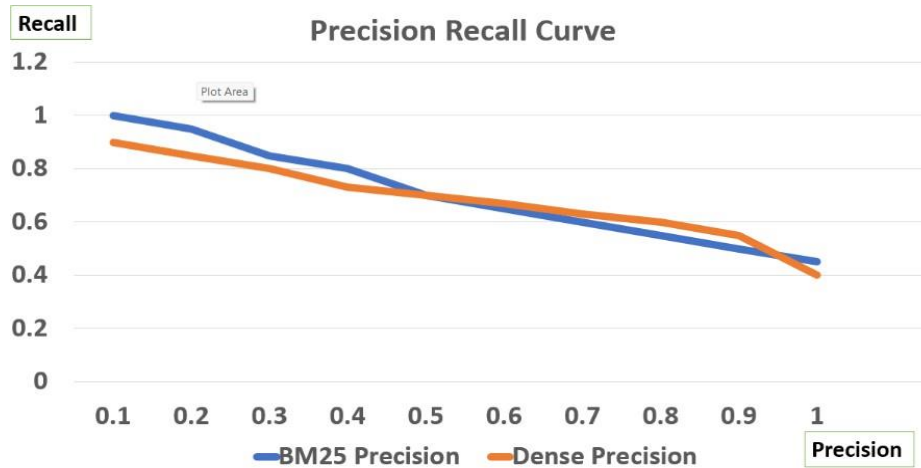


Fig. 3. Precision-Recall Curve Comparing BM25 and Dense Retrieval

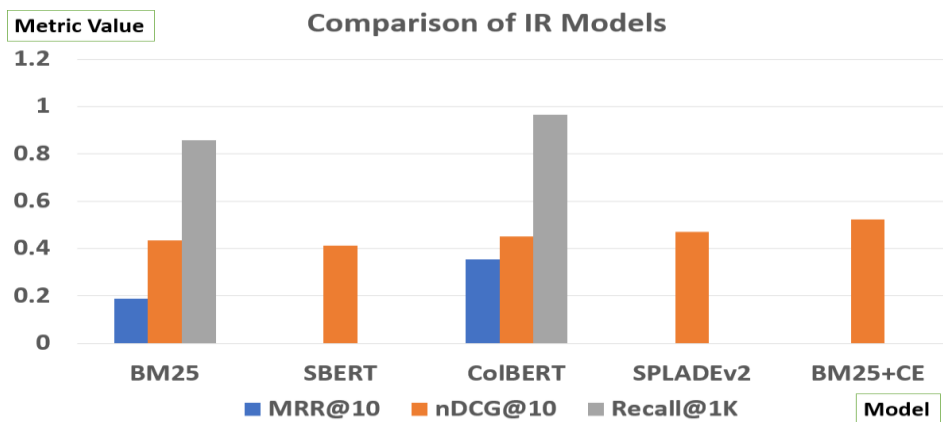


Fig. 4. Comparison of Information Retrieval Models

6. Open Challenges And Research Gaps

6.1 Scalability

Challenge: Storage and query processing are both taxed by billions of documents.

Current State: Ten billion documents are included in the MS MARCO Web Search dataset [19]. Scale degrades the approximate nearest neighbor search.

Gaps: Research on effective indexing structures for dense retrieval on a billion-scale scale is still ongoing. With minimal accuracy loss, static pruning can produce a speedup of 2-4× [20].

6.2 Multilingual Embeddings

Challenge: Low-resource languages are not well served by embedding models that were mostly trained on English.

Quantification: When compared to English, performance decreases by 20–40% for languages like Telugu, Swahili, or Tamil [6].

Gaps: Insufficient training data for languages with limited resources
Differences in script and morphology (such as agglutinative languages)
Terminology specific to a given domain and cultural setting
Improved cross-lingual transfer learning techniques are required.

6.3 Explainability

Challenge: Because neural ranking models are opaque, debugging and user trust are challenging.

Current Work: Anand et al. [21] survey explanation techniques including:

Attention visualization

LIME-based feature attribution

Probing classifiers

Counterfactual explanations

Gaps: Explanations are frequently inconsistent with model choices

Trade-off between model complexity and explanation quality

Absence of standardized explainability assessment measures

User research is required to evaluate the usefulness of explanations.

6.4 Domain Adaptation

Challenge: In specialized domains (including legal, medical, and scientific), models trained on general-domain data (like MS MARCO) perform poorly.

Example: In-domain training can boost nDCG@10 by 15–25 points over zero-shot models on BEIR's scientific datasets.

Gaps: Limited training data specific to a domain

During fine-tuning, catastrophic forgetting

Keeping domain-specific and generic knowledge in balance

Understudied few-shot adaptation methods

6.5 Efficiency vs. Accuracy Trade-offs

Challenge: More accurate models (cross-encoders, large LLMs) are prohibitively expensive for first-stage retrieval.

Current Approaches: Multi-stage ranking pipelines

Model compression (distillation, quantization, pruning)

Efficient architectures (late interaction, sparse neural)

Gaps: Systematic frameworks for optimizing cost-accuracy Pareto frontiers

Hardware-aware neural architecture search for IR

Energy efficiency considerations

6.6 Bias and Fairness

Challenge: Retrieval systems may amplify biases present in training data, leading to unfair or discriminatory outcomes.

Examples: Gender bias in search results

Racial bias in facial recognition-based image retrieval

Socioeconomic bias in recommendation systems

Gaps: Metrics for measuring retrieval fairness

Debiasing techniques that maintain effectiveness

Balancing relevance and fairness objective

6.7 Evaluation Limitations

Challenge: Standard benchmarks may not reflect real-world information needs.

Issues: Incomplete relevance judgments

Lexical bias in annotation (favoring BM25-like systems)

Narrow task definitions

Limited query diversity

Gaps: Need for more diverse, unbiased, and comprehensive evaluation frameworks.

7. Conclusion

This survey emphasizes the unique benefits of lexical, semantic, and hybrid retrieval techniques. Semantic models are better at capturing meaning and context than lexical models, despite lexical models' superior efficiency and accuracy. By combining these advantages, hybrid techniques like COIL and SPLADE outperform individual approaches, striking a balance between efficiency and accuracy in large-scale applications. However, there are still issues with multilingual retrieval, generalization, explainability, and scalability, particularly for low-resource languages. In order to guarantee that retrieval systems continue to be dependable and equitable while information continues to rise exponentially, future research is shifting toward adaptive hybrid systems, explainable neural models, and effective multilingual representations.

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